Yibo Yin

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EDUCATION

École polytechnique fédérale de Lausanne (EPFL) **Lausanne, Switzerland**
M.Sc. in Digital Humanities, School of Computer and Communication Sciences (IC) *09/2025 - Present*

- Coursework: Advanced Computer Graphics (6.0/6.0), Data Visualization (6.0/6.0), Applied Data Analysis (5.5/6.0)
- **GPA:** 5.59 /6.00

Wuhan University **Wuhan, China**
B.Eng. in Computer Science and Technology *09/2021 - 06/2025*

- **GPA:** 3.93 /4.00 (top 4%)

PUBLICATIONS

ArticuTable: Generating Instance-Level Interactive Rigid-Articulated 3D Tabletop Scenes from a Single Image
Kai Lv*, ***Yibo Yin****, Lijun Guo, Heng Fan, Kaihao Zhang, Xingping Dong
arXiv preprint, 2026
*Equal contribution.

MATStruct: High-Quality Medial Mesh Computation via Structure-aware Variational Optimization
Ningna Wang, Rui Xu, ***Yibo Yin***, Zichun Zhong, Taku Komura, Wenping Wang, Xiaohu Guo
ACM SIGGRAPH Asia 2025 (Conference Track)

RESEARCH EXPERIENCE

Image and Visual Representation Lab, EPFL *06/2026 - Present*
Research Assistant | PI: Prof. Sabine Süsstrunk; Supervisor: Zhuoqian Yang, Doctoral Assistant

- Investigating weight space generation for implicit neural representations, with the goal of learning modality-agnostic priors over neural fields across multiple data modalities.

Computer Graphics Lab, The University of Texas at Dallas *04/2024 - 11/2025*
Research Assistant (Remote) | Advisors: Prof. Xiaohu Guo and Dr. Ningna Wang

- Adapted Loop subdivision surface evaluation and sharp feature fitting algorithms to structured medial mesh.
- Designed a simplification algorithm for structured medial mesh to preserve geometric structure under subdivision.
- Proposed a subdivision surface fitting method for structured medial mesh, targeting sharp-feature preservation during fitting.

Graphics and Vision Lab, Wuhan University *05/2023 - 01/2024*
Research Assistant | Advisor: Prof. Chunxia Xiao

- Built an RGB-ToF/depth data acquisition pipeline and collected 10,000+ paired image-sensor samples.
- Proposed a novel monocular depth estimation method using RGB images and auxiliary ToF/depth signals.

SELECTED PROJECTS

Physically Based Ray Tracer *2026*
CS-440 Advanced Computer Graphics Rendering Competition (6.0/6.0), EPFL

- Extended the Nori framework with heterogeneous volumetric path tracing, delta and ratio tracking, Henyey-Greenstein phase functions, and MIS over emitter, phase, and BSDF sampling.
- Implemented HDR environment map lighting, thin-lens depth of field, motion blur, principled BSDF, and GGX rough dielectric materials.

AWARDS

- **Outstanding Student** (10% school-wide), Wuhan University *2022, 2023, 2024*
- **Second Class Scholarship** (10% school-wide), Wuhan University *2022*
- **Third Class Scholarship** (15% school-wide), Wuhan University *2023, 2024*
- **Lei Jun (Xiaomi) Computer Innovation and Development Fund**, Wuhan University *2024*

SKILLS

- **Languages:** Mandarin (Native), English (TOEFL iBT 102)
- **Programming Languages:** C++, C, Python, C#, GLSL, SQL, Verilog HDL, Java, LaTeX
- **Libraries / Frameworks / Tools:** PyTorch, Dr.Jit, Mitsuba, PBRT-v3, Blender, Eigen, CGAL, libigl, Git, CMake